MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, EHU-38/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for single-pointer, self-balancing potentiometer-type temperature indicator type EHU-38/A ac white lighted (12 pin receptacle).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-26500	Connector, General Purpose, Electrical, Miniature, Circular, Environmental Resisting General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-I-83262	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for
MIL-C-83723	Connector, Electrical, (Circular, Environment Resisting), Receptacles and Plugs, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

FSC 6685

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the requirements of MIL-I-83262 shall apply. In case of conflict between MIL-I-83262 and this specification, this specification shall govern.

3.2 Certification. The contractor shall certify that the indicators furnished under this specification have passed the qualification tests in accordance with MIL-I-83262.

3.2.1 Power consumption. The indicator steady-state power consumption shall not exceed 10Va exclusive of the lighting circuit.

3.2.2 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of at least 2000 hours at a 90 percent confidence level.

3.3 Design and construction. The indicator shall incorporate a self-balancing potentiometer-type circuit for temperature measurement. Other major components of the indicator are:

a. An input isolation-type power transformer

b. A dc torque motor.

3.3.1 Power supply. The power supply shall be in accordance with MIL-I-83262 for ac powered indicators.

3.4 Performance. The indicator shall meet the following requirements:

a. Scale error not exceeding the tolerances specified in table I

b. Friction error not exceeding 2°C over a temperature range of 100° to 1,200°C

c. Position error not exceeding 1°C

d. Case leakage not exceeding 0.1 micron cubic foot per hour

e. Warmup and pointer travel 100° to 1,000°C within 3 seconds at room temperature and 6 seconds at -55° and +100°C

f. Instantaneous indication change from 500° to 850°C in not more than 2 seconds at room temperature and 3 seconds at -55° and +100°C with any combination of voltages and frequencies from 95 to 135V and from 320 to 480 Hz

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g. When the signal is increased or decreased 0.08 mv a change of 2° ±1°C indication shall be noted at points between 600° and 850°C

h. Withstand temperatures ranging from -65° to +100°C

i. Withstand vibration of 5g from 5 to 500 Hz and as specified in MIL-I-83262

j. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for eight cycles with the ambient temperatures at 50°C

k. Indication change not exceeding 2°C when a leak resistance change from 5 to 200 ohms is made at specified test points between 200° and 1,000°C

l. Indication steady-state change not exceeding 2°C when either the chromel or alumel conductor is grounded separately at any point to the ground side (low potential side) of the 115V 400 Hz power supply.

m. Withstand the effects of beat frequencies and stray ac signals in accordance with MIL-I-83262

n. Withstand pressures ranging from 30 inches Hg down to 0.309 inch Hg (approximately sea level to 100,000 feet)

o. Withstand frequency variations ranging from 320 to 480 Hz, voltage variations from 95 to 135V steady-state and voltage surges to 175 ±5V.

p. Withstand eight cycles of alternate immersion in tap water at 85° ±2°C and 5° ±2°C

q. Withstand a mechanical shock of 10g for 11 ±1 milliseconds with no damage or malfunction of any component of the indicator.

r. Exposure to salt sea atmosphere for a period of 48 hours.

TABLE I. Test tolerances, EHU-38/A indicator

Test Points °C	Scale Error at Room Temperature ±°C	Scale Error at Low Temperature ±°C	Scale Error at High Temperature ±°C	Scale Error During Reliability Test ±°C	Friction Error ±°C
100	10	15	15	15	2
200	5	10	10	10	2
400	5	10	10	10	2
500	5	10	10	10	2
600	5	10	10	10	2
700	5	10	10	10	2
800	5	10	10	10	2
900	5	10	10	10	2
1,000	5	10	10	10	2
1,100	5	10	10	10	2
1,200	10	15	15	20	2

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3.5 Case. The indicator case shall be hermetically sealed in accordance with MS33639 for the 2-inch diameter size. The case length shall be 5.875 ±0.125 inches exclusive of the electrical receptacle. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein. The hermetic seal shall not be accomplished by use of materials that are affected by the action of any atmosphere to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than .03125 inch. The case shall be so designed that the internal mechanisms may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity.

3.6 Dial. The indicator dial shall be integrally lighted in accordance with MIL-L-27160. The dial shall have a minimum diameter of 1.75 inches as measured across the outer ends of the main dial graduation and shall be as shown on figure 1 and the requirements specified herein. This dial shall be linear and shall consist of a main dial and a subdial. The subdial shall have a diameter of .625 ±.0156 inch with scale graduations fabricated on the main dial plate. All graduations shall be linear within 0.5 of an angular degree. The scale arc of the main dial shall be 270 ±1 degrees. The subdial may be fabricated on a counterbored surface in the main dial plate or on a recessed disk beneath the level of the main dial plate, provided the viewing angle is at least as good as that of the main dial.

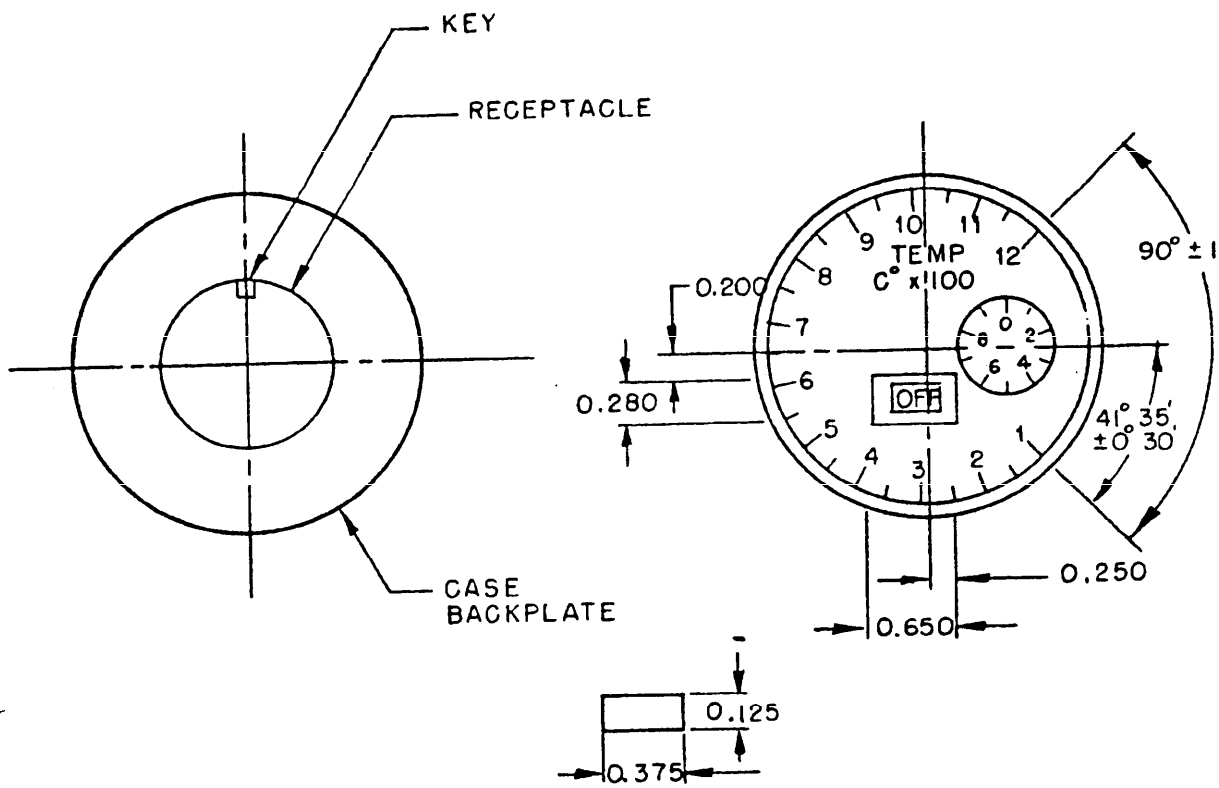
3.7 Pointers. The tips of the pointers shall extend to the inner ends of the respective major scale graduations ±0.016 inch.

3.7.1 Main dial pointers. The main dial pointer shall be in accordance with MS33585-8, except the width of the pointer tip shall be 0.020 ±0.005 inch. A minimum length of 0.60 inch of the main dial pointer shall be visible for all positions of the pointer. The pointer shall be integrally lighted in accordance with MIL-L-27160. Deviations to MS33585-8 will be permitted as required to accomplish the specified integral lighting.

3.7.2 Subdial pointer. The subdial pointer shall be as shown on figure 1. The entire length of the subdial pointer shall be lighted in accordance with MIL-L-27160. In lieu of a subdial pointer, an illuminated pointer configuration on a black rotating disk may be used, provided the lighting conforms to all requirements of MIL-L-27160. The length of the pointer is the distance from pointer tip to center of the pointer hub or pivot point.

3.8 Marking dimensions and colors. The dial and pointer colors shall be in accordance with FED-STD-595. The dimensions of the numerals and letters shall be as specified in table II.

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DIMENSIONS IN INCHES. TOLERANCE: ANGULAR ±0° 15

FIGURE 1. Dial and case backplate.

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TABLE II. Markings.

	Height or Length (Inch ± 0.005)	Width of Line or Graduation (Inch ± 0.005)	Color No.
Main dial numerals	0.156	0.031	37875
Main dial major graduations	.156	.031	37875
Main dial minor graduations	.094	.031	37875
Subdial numerals	.094	.020	37875
Subdial graduations	.062	.020	37875
Lettering: "TEMP" "°C x 100"	.094	.020	37875
"OFF"	.094	.020	17038
Pointers	---	---	37875
Background of dial	---	---	37038
Background of "OFF" warning	(See figure 1)	---	38905

3.9 Electrical receptacle. The indicator shall be provided with a hermetically sealed receptacle in accordance with MIL-C-83723, Series 3, type M83723-80H1212N. Pins 5 and 6 shall be made of alumel and chromel material respectively.

3.9.1 Indicator wiring. The electrical receptacle shall be wired as follows:

Pin	External Connection
1	115V, 400 Hz and 5V ground
5	Alumel conductor
6	Chromel conductor
8	Overtemperature warning switch
9	115V, 400 Hz input
11	5V AC or DC lighting input
12	Overtemperature warning switch

3.9.2 Grounding of case. The case shall be grounded by connecting a wire from the terminal of Pin 1 of the receptacle to a lug on the inner surface of the backplate of the indicator.

3.9.3 Output switch. The indicator shall include an isolated switch contact controlled by circuitry within the indicator. The switch point shall be internally adjustable over the temperature range of 821° to 835°C. The switch shall close with increasing temperature within the tolerances specified in table III. With decreasing temperature, the switch shall open when the temperature is 2 ±1°C below the set switch temperature. When 115 volt, 400 Hz power is not applied to the indicator, the switch shall be closed. Switch contact rating shall be 0.1 ampere minimum at 28 VDC.

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TABLE III. Accuracy of switch point.

Ambient temperature	Maximum Error
+25°C	±3°C
+38°C	±3°C
+100°C	±6°C
-54°C	±6°C

3.9.4 Power "OFF" warning. The indicator shall be provided with a power "OFF" warning as shown on figure 1. The warning shall display the word "OFF", with letters in gloss black, Color No. 17038, on a background of fluorescent red, Color No. 38905 of FED-STD-595. The warning shall remain in view until 115V 400 cycle power is applied for operation. When power is applied to the indicator, the warning shall move from view within 1 second and shall be replaced by a lusterless black surface. The warning shall be operated from the power supply of the indicator. The background of the warning shall be transilluminated when the warning is in view with power applied. This paragraph shall replace the requirements of 3.16, Open thermocouple circuit or power failure, of MIL-I-83262.

3.10 Excluded components. The design and fabrication of the indicator shall not include the following components:

a. Electromechanical choppers

b. Sleeve bearings

c. Plastic packaged integrated circuits.

3.11 Weight. The weight of the indicator shall not exceed 1.5 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, inspection and test

4.1.1 Sampling. Sampling shall be in accordance with MIL-I-83262 with the following additions: Add 4.1.4.9.1 to Individual tests of MIL-I-83262; also add 4.1.4.9.2 to Sampling plan B tests of MIL-I-83262.

4.1.2 Qualification inspection. Inspection for qualification shall be in accordance with MIL-I-83262 and as specified herein. Unless otherwise specified herein, the contractor shall perform the inspection and submit certification that the indicators conform to the requirements of this specification. A copy of the qualification test results shall be furnished to the Government.

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4.1.2.1 Certification. Certification is authorized for this item under the following conditions:

a. Contractor's responsibilities

(1) The contractor shall be responsible for performance of all inspections and tests necessary to insure that the item submitted for Government acceptance is in accordance with all the requirements of the drawings and specifications listed in the contract.

(2) Prior to shipment of the item, the contractor shall submit to the Government a certification of conformance based on either the contractor's inspection and tests, or those conducted by another qualified laboratory or inspection organization.

(3) The certification shall delineate all drawing and specification requirements being attested to, and shall be signed by a responsible officer of the contractor with authority to bind his principle.

(4) The certification shall contain the following statement: "I certify that the item described in this document has been inspected and tested and meets the following requirements. (Delineation of applicable requirements.)"

b. Government responsibilities. Item shall not be accepted until it has been properly identified with the certification. Government acceptance shall be made on the basis of the contractor's certification of conformance; however, the Government reserves the right to independently inspect items for conformance to all contractual drawings and specification requirements before acceptance.

4.1.3 Acceptance inspection. Acceptance inspection shall be in accordance with MIL-I-83262 except as otherwise specified herein.

4.1.4 Test methods. Unless otherwise specified herein, all test methods and procedures shall be in accordance with MIL-I-83262.

4.1.4.1 Power supply. Unless otherwise specified, the indicator shall be tested with 115V ±2V single-phase 400 ±5 Hz ac power with 4.5 ±0.5V ac or dc applied to the lighting circuit.

4.1.4.2 Electrical zero. NOT APPLICABLE

4.1.4.3 Scale error and friction. The friction error tests shall be in accordance with MIL-I-83262 and shall be inducted at 600°, 700°, 800°, and 900°C.

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4.1.4.4 Position error. This test shall be conducted at the 600°, 700° and 900°C test points.

4.1.4.5 Response time. Millivoltage signals and power shall be applied to provide a stable indication of 600°C. The millivoltage shall then be increased to produce an instantaneous change equivalent to 350°C. The time required for the indicator to fully respond to the 350°C change shall be noted and shall not exceed 2 seconds. This test shall be conducted using the following combinations of input voltage and frequencies: 115V 400 Hz, 95V 320 Hz, 95V 480 Hz, 135V 320 Hz and 135V 480 Hz. The test shall be repeated going from the high end of the expanded scale to the low end of the expanded scale. The same tolerance shall apply.

4.1.4.6 Sensitivity. The procedure shall be repeated at the 600° and 850°C test points.

4.1.4.7 Thermocouple lead resistance effect. This test shall be conducted at the 600°, 700°, and 800°C test points.

4.1.4.8 Grounding of thermocouple extension wires. Scale error shall be determined at the 600° and 700°C test points and the response time shall be determined in the 500° to 850°C range on increasing temperature.

4.1.4.9 Output switch. Operation of the output switch shall be tested using the following procedures as specified.

4.1.4.9.1 Output switch operation. Slowly vary the input thermocouple signal, both increasing and decreasing, to verify that with increasing temperature the switch closes at 828°C ±3°C or the set switch point and with decreasing temperature opens at 2°C ±1°C below the actual switch closing point.

4.1.4.9.2 Output switch function. Test function of the output switch at each of the switch set points (821°C, 828°C, 835°C) specified in table IV. For each set point and at each of the four ambient temperatures indicated verify with slowly increasing thermocouple input, that with increasing temperature, the switch closure occurs within the tolerances specified in table IV. Prior to each test at a specific temperature the indicator temperature shall be stabilized within the specified ambient for a period of at least 3 hours with power applied only 1 minute total on-time prior to test. This test can be combined with the applicable high and low temperature test specified in MIL-I-83262 when possible to shorten total test time.

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TABLE IV. Output switch test.

	821°C	828°C	835°C
Room Temp (+25°C)	±3°C	±3°C	±3°C
Aircraft Ambient Temp Test (+38°C)	±3°C	±3°C	±3°C
High Temp Test (+71°C)	±6°C	±6°C	±6°C
Low Temp Test (-54°C)	±6°C	±6°C	±6°C

4.1.4.10 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810C, equipment category b.1, procedure I, curve C. In addition to damaged described in MIL-I-83262, partial or erratic deflection of the power "OFF" warning during or after this test shall be cause for rejection.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-I-83262.

5.2 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.1 Nomenclature. The nomenclature shall be as follows:

INDICATOR, TEMPERATURE,
THERMOCOUPLE, SELF-BALANCING
POTENTIOMETER EHU-38/A

6. NOTES

6.1 All notes specified in MIL-I-83262 are applicable to this specification.

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Reviewer activity:
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Preparing activity:
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